

**Sri Lanka Institute of Information
Technology**



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Web-Based Supermarket Ordering System

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01.Introduction

1.1 Project Overview

The web-based supermarket ordering system is a digital platform designed to automate and integrate the core operations of a supermarket, while also modernizing and enhancing customer experience. The system allows customers to shop online with real-time order tracking while integrating key supermarket functions like product management, inventory control, payments, delivery coordination, and marketing. This project aims to address the challenges by delivering a streamlined platform that enhances operational efficiency, enables real-time communication and elevates customer satisfaction.

The system is designed to include six main functional areas such as user management, product and inventory management, order processing management, payment management, delivery management and marketing and promotion management. Each module is developed independently by each member of the group enabling parallel testing and easy updates without affecting the overall system. To ensure that the system meets real-world needs, Microsoft Copilot bot was used in the requirement gathering phase. It assisted simulating stakeholders' interactions, uncovering domain specific insights and identifying common industry challenges. These insights were used to develop detailed requirements for each module, with a focus on addressing the real challenges faced in the supermarket. The system employs role-based access control to ensure that every user—be it an admin officer, inventory manager, delivery coordinator, marketing manager, regular customer, or payment merchant representative—has clearly defined permissions and responsibilities.

By integrating all core operations into one platform, the web-based supermarket ordering system improves service delivery, enhances the shopping experience, ensures data security and enables staff to manage their duties effectively through real-time data access and automated workflows.

1.2 Objectives

Primary Objectives:

- Provide a user-friendly environment for customers to browse products, place orders and make payments easily from anywhere.
- Improve inventory control by enabling real-time tracking and management of stock levels to prevent overstocking and stockouts.
- Provide a role-based access control system to ensure secure and user-specific functionalities for admin, staff and customers.

Technical Objectives:

- Follow the full SDLC, from requirement gathering to testing and development.
- Develop using Java, Spring Boot, Microsoft SQL and responsive front-end technologies (HTML/CSS/JS).
- Design an efficient database scheme to manage data related to users, products, transactions and deliveries efficiently.

1.3 Stakeholders/Users

- Admin Officer
- Inventory Manager
- Regular Customer
- Marketing Manager
- Payment Manager
- Delivery Coordinator

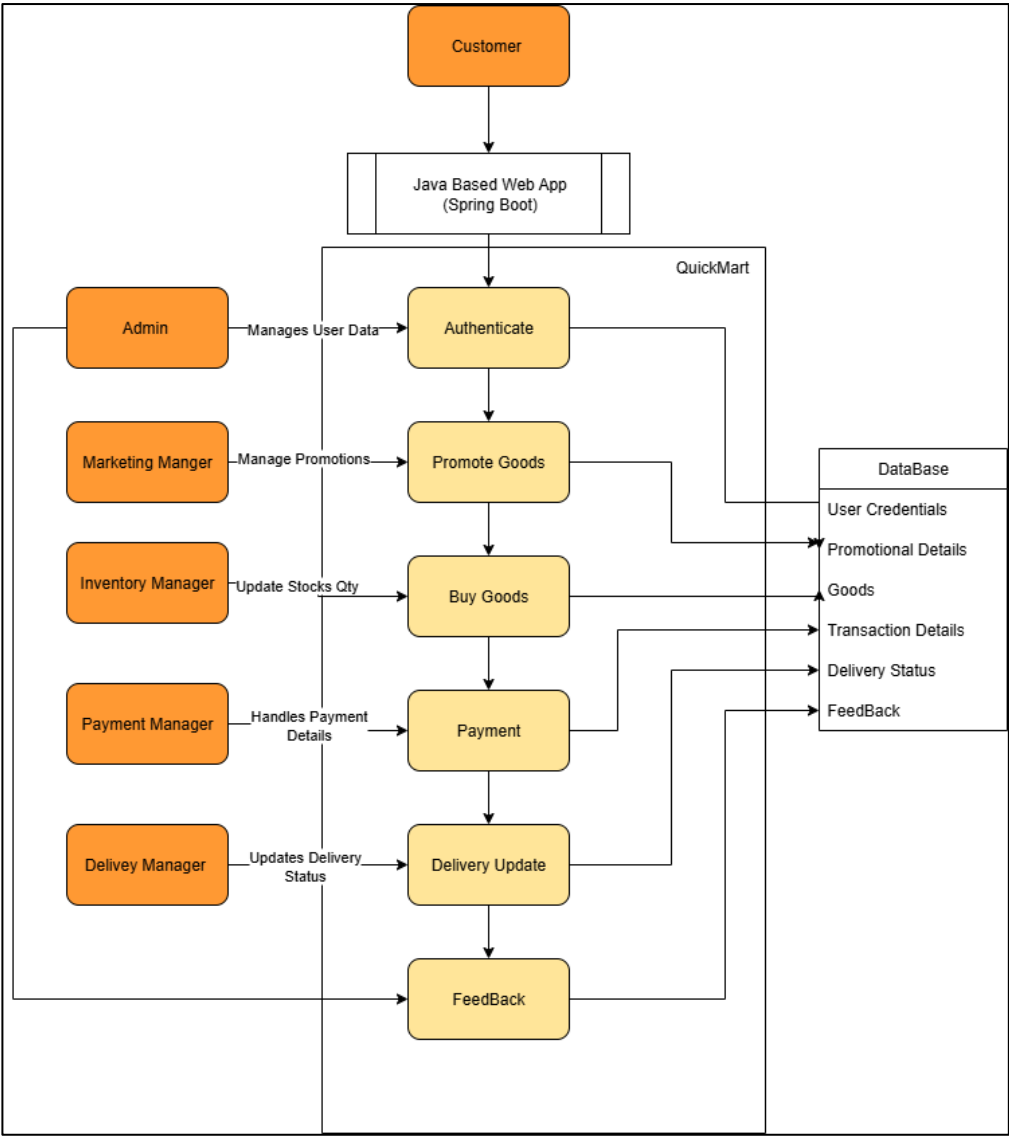
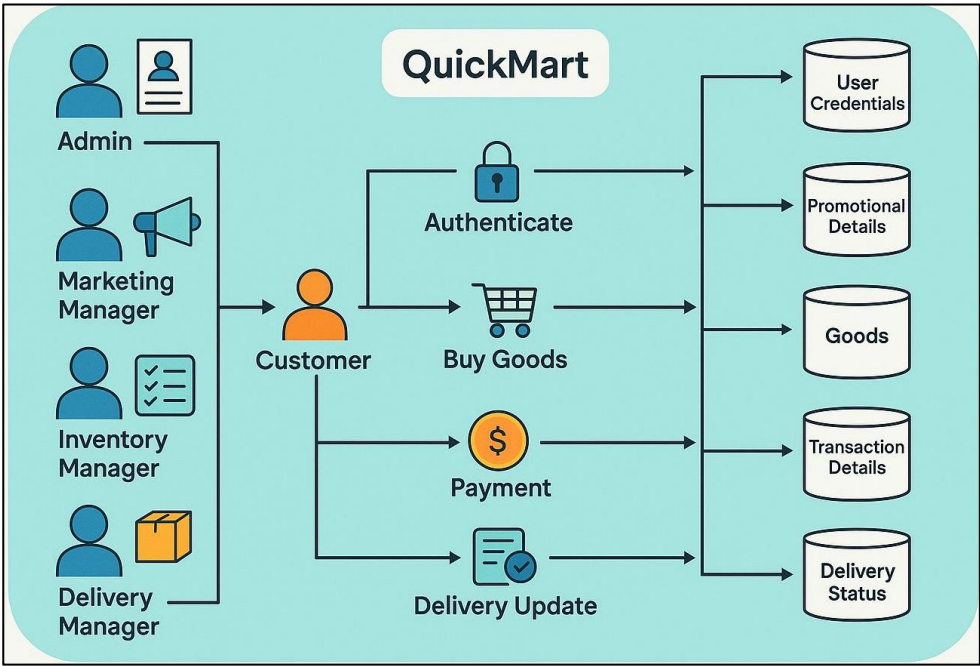
1.4 Scope and Limitations**Scope:**

- Allow customers to browse products, place orders, make payments, and view order status online.
- Manage inventory, product catalog, and stock updates in real time.
- Provide role-based access for admin, staff and customers.
- Developed as a web-based system using Java, Spring Boot, and Microsoft SQL.

Limitations:

- Available only as a web application (No mobile version).
- Real-time delivery tracking (e.g., GPS-based) is not implemented; only status updates are supported.
- Limited language support (English only).
- Supports only card payments and cash on delivery, no bank transfer option.

03. System Overview Diagram



04. Functional Requirements

1. User management

- Create and manage user accounts.
- Give different access levels to staff.
- View reports like orders, payments, and deliveries.
- Block or fix accounts if there are problems.
- Update system settings when needed.

2. Inventory Management

- Add new items to the system
- Update item details like price, quantity, and description
- Get alerts when stocks are low
- Remove items that are no longer available
- View reports of fast-moving and slow-moving items

3. Order Processing System

- Customers can browse, search, filter, and view products.
- Add/remove products to/from shopping cart or Wishlist.
- Place orders with delivery and promo options.
- View, reorder, or cancel orders from order history.
- Provide real-time order tracking.

4. Marketing and Promotion Management

Can create/update/delete discounts, promotions, campaigns.

- Send personalized notifications/emails based on customer behavior.
- Manage campaign performance
- Generate reports and item popularity

5. Payment Management

- Integrate with third-party payment gateways (e.g., Stripe, PayPal).
- Support multiple payment methods (cards, wallets, COD).
- Generate receipts and notify customers after payment.
- Alert on failed payments and provide retry/refund options.
- Generating reports to track daily and monthly reports

6. Delivery Management

- Delivery coordinator assigns orders to delivery staff with estimated delivery slots.
- Update delivery statuses: packed, in transit, delivered.
- Send updates to customers at key stages.
- Log delivery confirmations and issues (delays, failed deliveries).

05. Non-Functional Requirements

Performance & Scalability

- Support at least 500 concurrent users with fast response time.
- Handle minimum 10,000 products without performance issues.

Security

- Encrypt data in transit (HTTPS) and at rest (hashed passwords).
- Implement role-based access control.
- Log security events for auditing.

Availability & Reliability

- Achieve at least 99.5% uptime, excluding planned maintenance.
- Implement automatic recovery for failed components.

Maintainability & Extensibility

- Follow modular architecture for easy future feature integration.
- Support automated testing and deployment.

Usability

- Ensure mobile-responsive, user-friendly interface.
- Allow new users to complete key tasks (browse, order, pay) without training.

Compliance

- Comply with data privacy regulations.
- Maintain transaction records meeting local tax/reporting requirements.

06. Major Stakeholders

Stakeholder	Description
Admin Officer	- Manages user accounts and access. - Check reports and monitor user activity.
Inventory Manager	- Check and update Stock levels. - Update item quantities. - Update product prices.
Regular Customer	- Browse products. - Place orders. - Make payments and provide feedback.
Marketing Manager	- Manage product advertisements. - Provide discounts and seasonal deals to customers.
Payment Manager	- Ensure that payment services are secure and check if payments are successful or failed. - Confirm the customer payment details are handled securely.
Delivery Coordinator	- Check new orders. - Assign orders for delivery. - Monitor deliveries. - Ensure the orders are placed and update the delivery reports.

07. Six Major Functions

Function	What it does	Who will use it	Sub-functions	Expected outcome
User Management (Jayalath J.G.H)	Manage user accounts and permissions	Admin Officer	• Create/update/delete accounts • Assign roles • View reports (orders payments, deliveries) • Update system settings	Secure, role-based access, accurate reporting, system integrity
Product & Inventory Management (Alahakoon B.M.B.S)	Manage product listings and stock levels	Inventory manager	• Add/ update/ remove products • Monitor low stock alerts • View item movement reports	Accurate inventory, product availability, low stock alerts
Order Processing System (Fernando S.A.W)	Browse, shop, place and manage orders	Regular customer	• Browse/search/filter products • Place/cancel/reorder • Track orders	Smooth customer shopping experience, real-time updates
Payment Management (Sanvidu M.G.M)	Process and track secure payments	Payment Manager	• Integrate with payment gateways • Accept multiple payment methods • Generate receipts • Handle failed payments and refunds	Secure transactions, reliable payment history, refund handling
Delivery Management (Dinod P.A.S)	Assign and track deliveries, notify customers	Delivery coordinator	• Assign delivery staff • Update delivery staff • Send delivery notifications	Timely and trackable deliveries, better customer satisfaction
Marketing and Promotion Management (Thahani N)	Create and manage campaigns, promotions	Marketing Manager	• Add/edit/delete campaigns • Send personalized notifications • View item popularity reports	Increased customer engagement and sales, performance tracking

08. Minor Functions

Authentication and Authorization

- Login and logout functionality for users and staff
- Password reset
- Two-factor authentication (optional)
- Role-based access control

Search and Filter

- Product search by name, category, brand, etc.
- Filter by price, popularity, ratings
- Sort by newest, price, best-sellers

User Profile Management

- Update personal details (name, email, address, password)
- Manage saved delivery addresses

System Settings

- Set tax rates, currency, delivery charges
- Define promotional rules (e.g., buy 1 get 1)
- Configure order cancellation rules or refund timelines

Responsive UI

- Adaptive design for mobile, tablet, and desktop views
- Fast and accessible front-end experience

Audit and Logs

- Log user and admin actions
- Track changes made to orders, inventory, or settings

09. System Limitations

Feature Limitations

- Real-time GPS-based delivery tracking is not implemented. Customers receive only status updates (e.g., packed, in transit, delivered).
- Multi-language support is not available; the system operates in English only.

Payment Method Constraints

- The system supports only credit/debit card payments and cash on delivery.
- Bank transfers, digital wallets (e.g., PayPal, eZ Cash), and crypto payments are not supported at this stage.

Platform Limitations

- The system is available only as a web application; there is no dedicated mobile app.
- The web interface is optimized for modern desktop and mobile browsers, but not for native mobile platforms (iOS/Android).

Device & Access Constraints

- It is not optimized for offline access. Internet connectivity is required for all functionalities.

10. Simple Timeline

	Week											
Milestones	3	4	5	6	7	8	9	10	11	12	13	14
Project Scope & Requirements Gathering												
Project Scope & Requirements Finalization												
System Architecture and Database Design												
Task assignment, set up development environment												
Core functionality development												
Module testing												
Cart, Order & Customer management												
Integration testing												
Payment & Feedback module development												
Complete system integration testing												
Promotions & Final Integrations, Bug fixing & system testing												
Final testing, documentation and presentation												

11. Conclusion

Web-Based Supermarket Ordering System is meant to make supermarket operations easier and more efficient. It brings everything like managing users, tracking inventory, managing orders, payment, delivery, and promotions into one simple online system. Using technologies like Java, Spring Boot, and Microsoft SQL, it facilitates customers with an easy shopping experience through real-time browsing, ordering, tracking, wish lists, and easy payment options. As it does that, it helps employees minimize mistakes and save time using automated procedures. It's safe, can handle big groups of users and products, and can grow with additions like mobile apps and GPS tracking of location.

The team has developed a good 14-week plan to achieve this. Prior to that, they will have accomplished what the system will demand and drawn the database out. Then, during weeks 7 to 12, every team member will build different pieces like inventory or user management while also signing in each week to make sure they're working properly. The final few weeks will be for putting everything together, testing thoroughly, bug fixing, and getting everything ready for the ultimate demo.

In short, this system is a judicious step towards going online with supermarkets. With good planning, coordination, and focus on the users of this system, the team is bound to develop something functional, reliable, and future proof.